COMPUTER ENGINEERING
QUEZON CITY

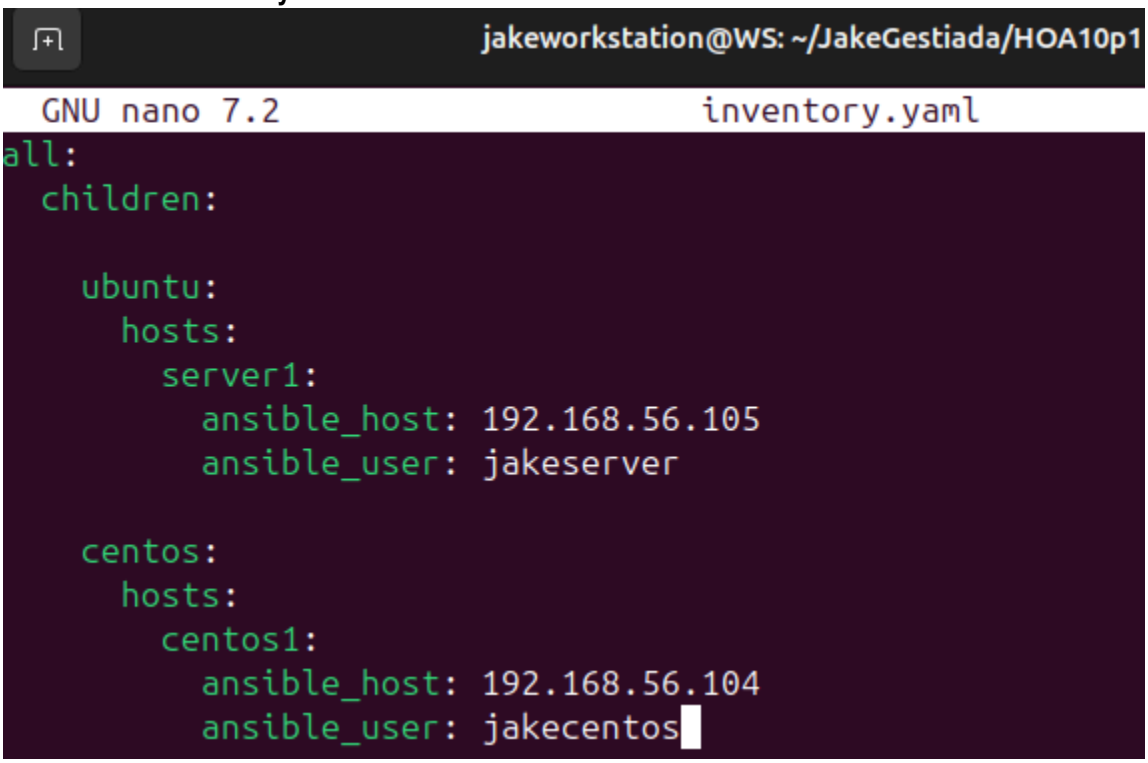
| | | | |
|---|---|---|---|
| **COURSE CODE/TITLE:** | CPE 212- Automating Server Management | **SECTION:** | CPE31s2 |
| **DATE PERFORMED:** | 04/10/2026 | **DATE SUBMITTED:** | 06/10/2026 |
| **NAME:** | Damien Jake Gestiada | **INSTRUCTOR:** | Engr. Jimlord Quejado |

**ACTIVITY NO. 10** **Install, Configure and Manage Enterprise Log Monitoring via Ansible**

## 1. PROCEDURE OUTPUT

1. Create a playbook that:
a. Install and configure Elastic Stack in separate hosts (Elastic Search, Kibana, Logstash)
2. Apply the concept of creating roles.
3. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)
4. Show an output of the installed Elastic Stack for both Ubuntu and CentOS.
5. Make sure to create a new repository in GitHub for this activity.

**Task 1: Create the Inventory File**




- This task creates the inventory.yaml file that lists the managed servers. The Ubuntu Server1 uses IP

192.168.56.105, while the CentOS server uses IP 192.168.56.104, allowing the Control Node to connect to both machines using Ansible.

**Task 2: Create the Main Playbook**

```
 GNU nano 7.2                                                site.yml
---
- name: Install Elasticsearch and Kibana on Ubuntu
  hosts: ubuntu
  become: true
  roles:
    - elasticsearch
    - kibana

- name: Install Logstash on CentOS
  hosts: centos
  become: true
  roles:
    - logstash
```

- This task creates site.yml, which controls the installation of the Elastic Stack components. Elasticsearch and Kibana are assigned to the Ubuntu server, while Logstash is assigned to the CentOS server.

**Task 3: Create the Ansible Roles Structure**

```
jakeworkstation@WS:~/JakeGestiada/HOA10p1$ tree
.
├── inventory.yaml
├── roles
│   ├── elasticsearch
│   │   └── tasks
│   │       └── main.yml
│   ├── kibana
│   │   └── tasks
│   │       └── main.yml
│   └── logstash
│       └── tasks
│           └── main.yml
└── site.yml
```

## Task 4: Create the Elasticsearch Ansible Role

```
 GNU nano 7.2                                    roles/elasticsearch/tasks/main.yml
---
# 1. Install prerequisites
- name: Install Elasticsearch prerequisites
  ansible.builtin.apt:
    name:
      - apt-transport-https
      - wget
      - curl
      - gnupg
    state: present
    update_cache: true

# 2. Configure Elasticsearch system settings
- name: Configure vm.max_map_count
  ansible.builtin.copy:
    dest: /etc/sysctl.d/99-elasticsearch.conf
    owner: root
    group: root
    mode: "0644"
    content: |
      vm.max_map_count=1048576

- name: Apply vm.max_map_count
  ansible.builtin.command:
    cmd: sysctl -w vm.max_map_count=1048576
  changed_when: false

# 3. Add Elastic signing key
- name: Download Elastic signing key
  ansible.builtin.get_url:
```

```
jakeworkstation@WS: ~/JakeGestiada/HOA10p1/roles/elasticsearch/tasks
 GNU nano 7.2                                    main.yml
  changed_when: false

# 3. Add Elastic signing key
- name: Download Elastic signing key
  ansible.builtin.get_url:
    url: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    dest: /tmp/GPG-KEY-elasticsearch
    mode: "0644"

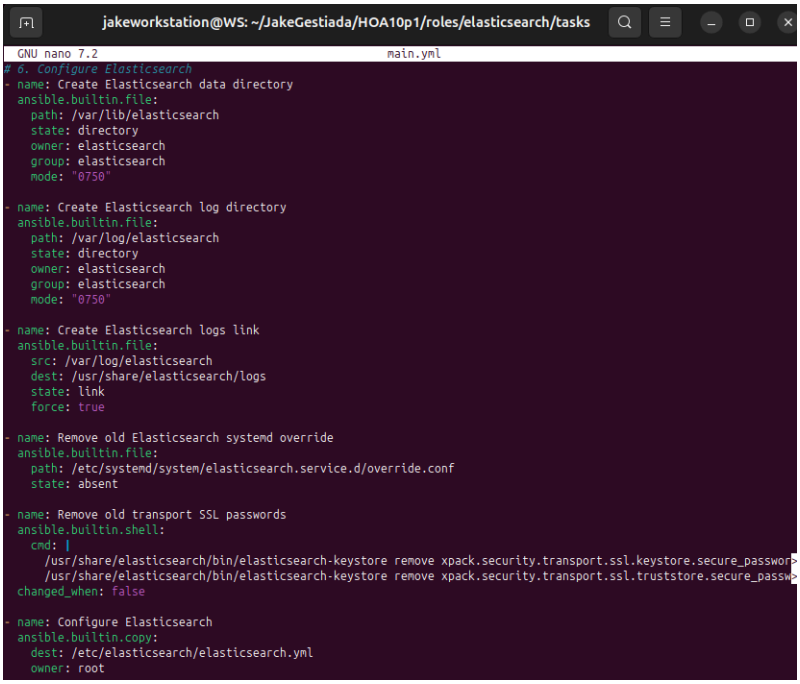
- name: Convert Elastic signing key
  ansible.builtin.command:
    cmd: gpg --dearmor -o /usr/share/keyrings/elasticsearch-keyring.gpg /tmp/GPG-KEY-elasticsearch
  args:
    creates: /usr/share/keyrings/elasticsearch-keyring.gpg

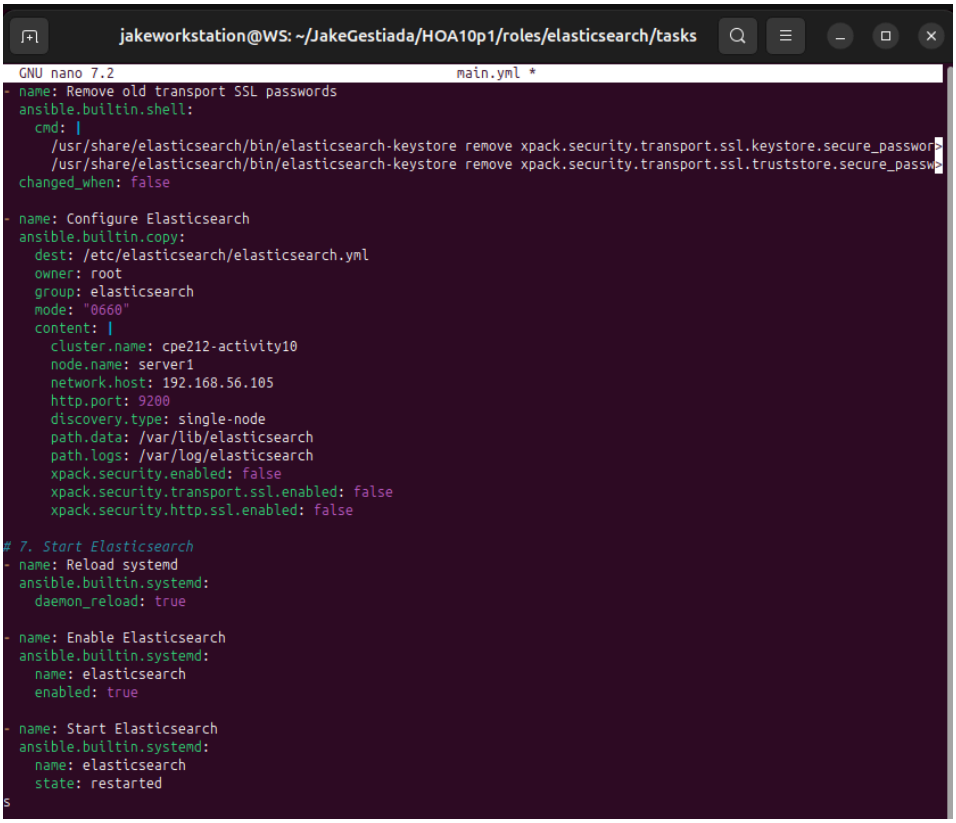
# 4. Add Elastic repository
- name: Add Elastic 9.x repository
  ansible.builtin.apt_repository:
    repo: "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg] https://artifacts.elastic.co/packages/9.x/ap
    filename: elastic-9.x
    state: present
    update_cache: true

# 5. Install Elasticsearch
- name: Install Elasticsearch
  ansible.builtin.apt:
    name: elasticsearch
    state: present

# 6. Configure Elasticsearch
- name: Create Elasticsearch data directory
  ansible.builtin.file:
    path: /var/lib/elasticsearch
    state: directory
    owner: elasticsearch
    group: elasticsearch
    mode: "0750"

- name: Create Elasticsearch log directory
  ansible.builtin.file:
```




- This task creates the Elasticsearch role that adds the Elastic signing key and repository, installs Elasticsearch, and prepares its required directories and configuration. It automates the setup of Elasticsearch on the Ubuntu Server1 managed node.

## Task 5: Create the Kibana Ansible Role

```
  GNU nano 7.2                                    roles/kibana/tasks/main.yml
---
# 1. Install prerequisites
- name: Install Kibana prerequisites
  ansible.builtin.apt:
    name:
      - apt-transport-https
      - wget
      - curl
      - gnupg
    state: present
    update_cache: true

# 2. Add Elastic signing key
- name: Download Elastic signing key
  ansible.builtin.get_url:
    url: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    dest: /tmp/GPG-KEY-elasticsearch
    mode: "0644"

- name: Convert Elastic signing key
  ansible.builtin.command:
    cmd: gpg --dearmor -o /usr/share/keyrings/elasticsearch-keyring.gpg /tmp/GPG-KEY-elasticsearch
  args:
    creates: /usr/share/keyrings/elasticsearch-keyring.gpg

# 3. Add Elastic repository
- name: Add Elastic 9.x repository
  ansible.builtin.apt_repository:
    repo: "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg] https://artifacts.elastic.co/packages/9.x/apt stable main"
    filename: elastic-9.x
```

```
jakeworkstation@WS: ~/JakeGestiada/HOA10p1/roles/kibana/tasks
  GNU nano 7.2                                    main.yml
- name: Add Elastic 9.x repository
  ansible.builtin.apt_repository:
    repo: "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg] https://artifacts.elastic.co/packages/9.x/ap>
    filename: elastic-9.x
    state: present
    update_cache: true

# 4. Install Kibana
- name: Install Kibana
  ansible.builtin.apt:
    name: kibana
    state: present

# 5. Configure Kibana
- name: Configure Kibana
  ansible.builtin.copy:
    dest: /etc/kibana/kibana.yml
    owner: root
    group: kibana
    mode: "0660"
    content: |
      server.port: 5601
      server.host: "192.168.56.105"
      elasticsearch.hosts: ["http://192.168.56.105:9200"]

# 6. Start Kibana
- name: Reload systemd
  ansible.builtin.systemd:
    daemon_reload: true

- name: Enable Kibana
  ansible.builtin.systemd:
    name: kibana
    enabled: true

- name: Start Kibana
  ansible.builtin.systemd:
    name: kibana
    state: restarted
```

- This task creates the Kibana role that installs the required packages, adds the Elastic repository, installs Kibana, and configures it to connect to Elasticsearch on 192.168.56.105:9200. It also enables and starts the Kibana service on Ubuntu Server1.

## Task 6: Create the Logstash Ansible Role

GNU nano 7.2 roles/logstash/tasks/main.yml

```yaml
---
# 1. Install prerequisites
- name: Install Logstash prerequisites
  ansible.builtin.package:
    name:
      - curl
      - wget
    state: present

# 2. Add Elastic signing key
- name: Import Elastic signing key
  ansible.builtin.rpm_key:
    key: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    state: present

# 3. Add Elastic repository
- name: Add Elastic 9.x repository
  ansible.builtin.yum_repository:
    name: elastic-9.x
    description: Elastic repository for 9.x packages
    baseurl: https://artifacts.elastic.co/packages/9.x/yum
    gpgcheck: true
    gpgkey: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    enabled: true
    state: present

# 4. Install Logstash
- name: Install Logstash
  ansible.builtin.package:
    name: logstash
```

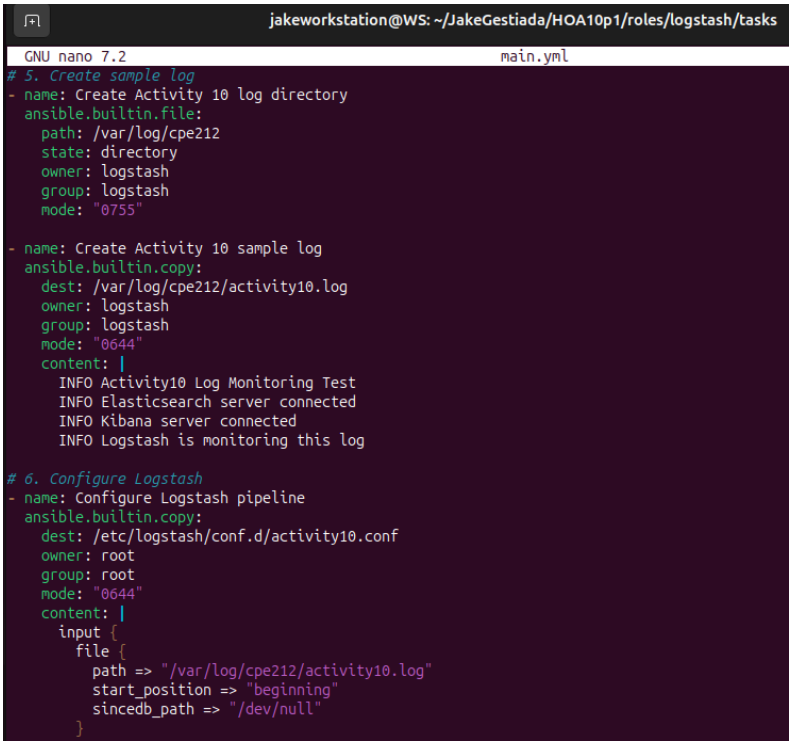
jakeworkstation@WS: ~/JakeGestiada/HOA10p1/roles/logstash/tasks

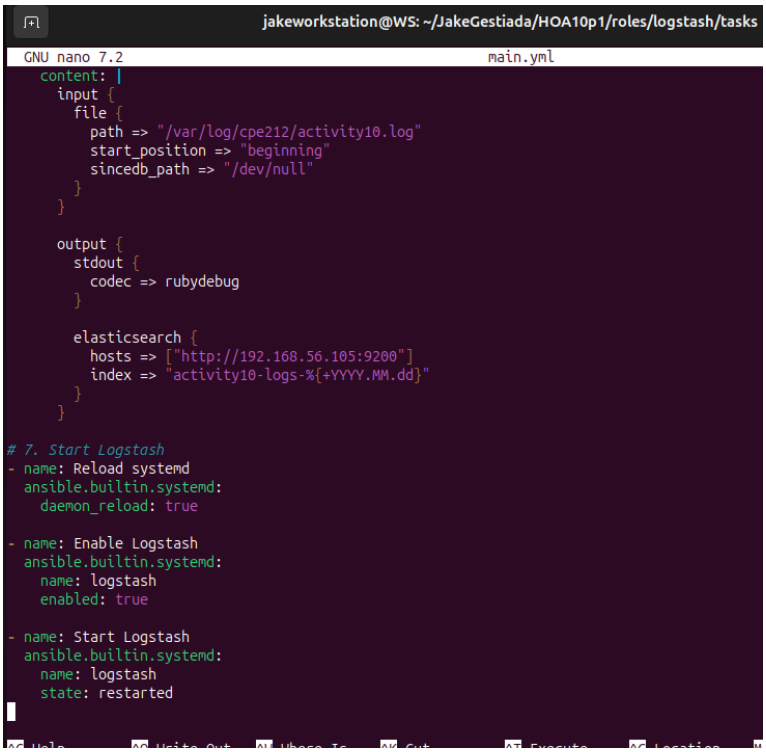
GNU nano 7.2 main.yml

```yaml
# 5. Create sample log
- name: Create Activity 10 log directory
  ansible.builtin.file:
    path: /var/log/cpe212
    state: directory
    owner: logstash
    group: logstash
    mode: "0755"

- name: Create Activity 10 sample log
  ansible.builtin.copy:
    dest: /var/log/cpe212/activity10.log
    owner: logstash
    group: logstash
    mode: "0644"
    content: |
      INFO Activity10 Log Monitoring Test
      INFO Elasticsearch server connected
      INFO Kibana server connected
      INFO Logstash is monitoring this log

# 6. Configure Logstash
- name: Configure Logstash pipeline
  ansible.builtin.copy:
    dest: /etc/logstash/conf.d/activity10.conf
    owner: root
    group: root
    mode: "0644"
    content: |
      input {
        file {
          path => "/var/log/cpe212/activity10.log"
          start_position => "beginning"
          sincedb_path => "/dev/null"
        }
```




- This task creates the Logstash role for the CentOS managed node, including the Elastic repository, Logstash installation, sample log file, and monitoring pipeline. It also configures Logstash to send the collected logs to Elasticsearch at 192.168.56.105:9200.

## Task 7: Run the Ansible Playbook

```
jakeworkstation@WS:~/JakeGestiada/HOA10p1$ ansible-playbook -i inventory.yaml site.yml --limit ubuntu --ask-become-pass
BECOME password:

PLAY [Install Elasticsearch and Kibana on Ubuntu] ******************************************************************************

TASK [Gathering Facts] *********************************************************************************************************
ok: [server1]

TASK [elasticsearch : Install Elasticsearch prerequisites] *********************************************************************
ok: [server1]

TASK [elasticsearch : Configure vm.max_map_count] ******************************************************************************
ok: [server1]

TASK [elasticsearch : Apply vm.max_map_count] **********************************************************************************
ok: [server1]

TASK [elasticsearch : Download Elastic signing key] ****************************************************************************
ok: [server1]

TASK [elasticsearch : Convert Elastic signing key] *****************************************************************************
ok: [server1]

TASK [elasticsearch : Add Elastic 9.x repository] ******************************************************************************
ok: [server1]

TASK [elasticsearch : Install Elasticsearch] ***********************************************************************************
ok: [server1]

TASK [elasticsearch : Create Elasticsearch data directory] *********************************************************************
ok: [server1]

TASK [elasticsearch : Create Elasticsearch log directory] **********************************************************************
```

```
TASK [kibana : Install Kibana prerequisites] ***********************************************************************************
ok: [server1]

TASK [kibana : Download Elastic signing key] ***********************************************************************************
ok: [server1]

TASK [kibana : Convert Elastic signing key] ************************************************************************************
ok: [server1]

TASK [kibana : Add Elastic 9.x repository] *************************************************************************************
ok: [server1]

TASK [kibana : Install Kibana] *************************************************************************************************
changed: [server1]

TASK [kibana : Configure Kibana] ***********************************************************************************************
changed: [server1]

TASK [kibana : Reload systemd] *************************************************************************************************
ok: [server1]

TASK [kibana : Enable Kibana] **************************************************************************************************
changed: [server1]

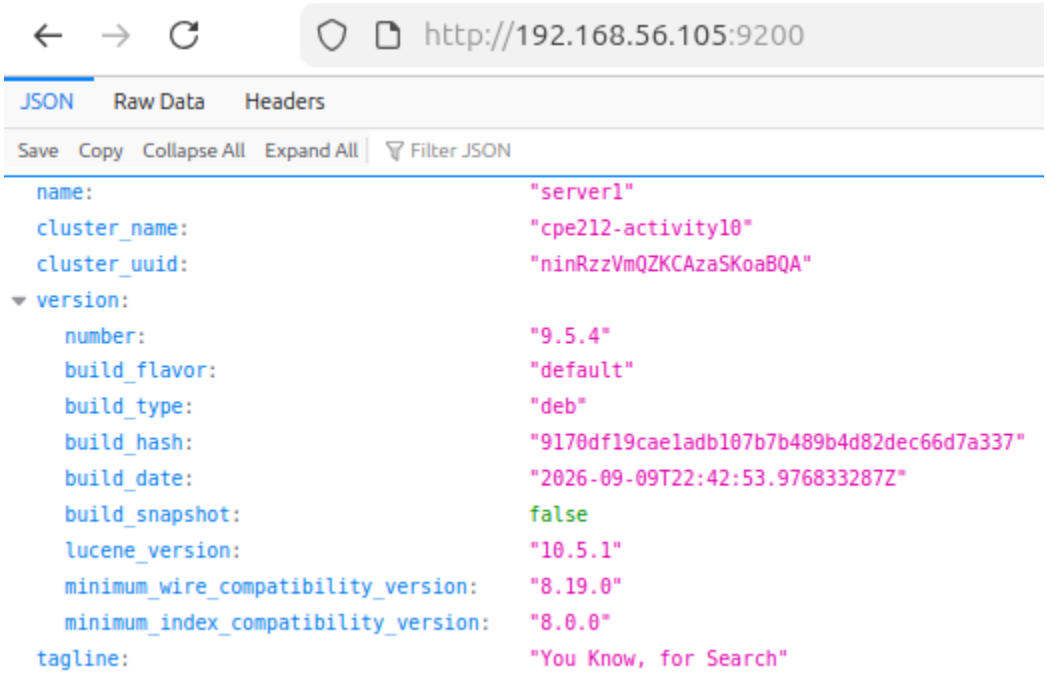
TASK [kibana : Start Kibana] ***************************************************************************************************
changed: [server1]

PLAY [Install Logstash on CentOS] **********************************************************************************************
skipping: no hosts matched

PLAY RECAP *********************************************************************************************************************
server1                    : ok=26   changed=6    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

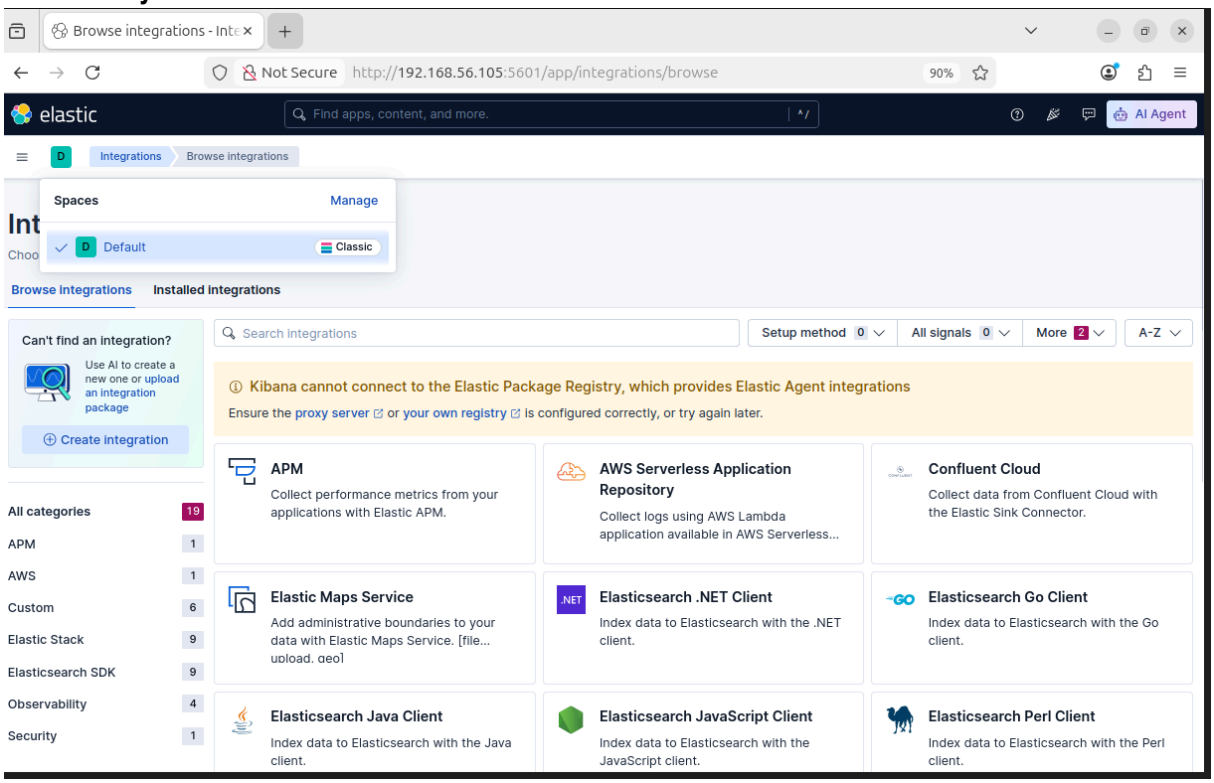
- This task runs site.yml using Ansible with --limit ubuntu, applying the Elasticsearch and Kibana roles to the Ubuntu Server1. The output shows the installation and configuration tasks being executed successfully.

**Task 8: Verify Elasticsearch in the Web Browser**




- This task verifies that Elasticsearch is running correctly by opening http://192.168.56.105:9200 in Firefox. The displayed JSON information confirms that Elasticsearch on Server1 is active and responding.

## Task 9: Verify Kibana in the Web Browser




- This task verifies that Kibana is running on Ubuntu Server1 by opening its web interface at http://192.168.56.105:5601. The displayed Kibana interface confirms that the Kibana service is accessible and ready to work with Elasticsearch.

## Task 10: Install and Configure Logstash on CentOS

```
jakeworkstation@WS:~/JakeGestiada/HOA10p1$ ansible-playbook -i inventory.yaml site.yml --limit centos --ask-become-pass
BECOME password:

PLAY [Install Elasticsearch and Kibana on Ubuntu] ******************************************************************************************
skipping: no hosts matched

PLAY [Install Logstash on CentOS] **********************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************
ok: [centos1]

TASK [logstash : Install Logstash prerequisites] *******************************************************************************************
ok: [centos1]

TASK [logstash : Import Elastic signing key] ***********************************************************************************************
changed: [centos1]

TASK [logstash : Add Elastic 9.x repository] ***********************************************************************************************
changed: [centos1]

TASK [logstash : Install Logstash] *********************************************************************************************************
changed: [centos1]

TASK [logstash : Create Activity 10 log directory] *****************************************************************************************
changed: [centos1]

TASK [logstash : Create Activity 10 sample log] ********************************************************************************************
changed: [centos1]

TASK [logstash : Configure Logstash pipeline] **********************************************************************************************
changed: [centos1]

TASK [logstash : Reload systemd] ***********************************************************************************************************
ok: [centos1]

TASK [logstash : Enable Logstash] **********************************************************************************************************
changed: [centos1]
```

```
TASK [logstash : Enable Logstash] ***********************************************************************************************************
changed: [centos1]

TASK [logstash : Start Logstash] ************************************************************************************************************
changed: [centos1]

PLAY RECAP **********************************************************************************************************************************
centos1                    : ok=11   changed=8    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

- This task runs the Ansible playbook on the CentOS managed node to install and configure Logstash. The result shows all Logstash tasks completed successfully with failed=0 and unreachable=0.

**Task 11: Verify Logstash Service Status**

```
jakeworkstation@WS:~/JakeGestiada/HOA10p1$ ansible centos -i inventory.yaml -b -m shell -a "systemctl is-active logstash" --ask-become-pass
BECOME password:
centos1 | CHANGED | rc=0 >>
active
```

- This task checks the Logstash service on the CentOS managed node using Ansible and systemctl. The result shows active, confirming that Logstash is running properly.

**Task 12: Confirm Logstash is Active**

```
jakeworkstation@WS:~/JakeGestiada/HOA10p1$ ansible centos -i inventory.yaml -b -K -m shell -a "systemctl is-active logstash"
BECOME password:
centos1 | CHANGED | rc=0 >>
active
```

- This task checks the Logstash service on the CentOS managed node using Ansible. The active result confirms that Logstash is currently running successfully.

## 2. ANSWERS TO SUPPLEMENTAL ACTIVITY

**Reflections:**

Answer the following:

1. What are the benefits of having log monitoring tool?
   - A log monitoring tool helps in monitoring important events and activities in the servers, applications, and networks. It has the capability to identify any critical event and alert us right away when there is an issue. This tool greatly helps in narrowing down and finding the roots of performance and security problems more quickly. Through observing the logs we can spot issues before they cause any serious downtime. It contributes to the enhancement of system reliability, security, management of risk, among others.

**Conclusion**

- This activity showed us how Ansible might be one of the tools to automate installation and configure log monitoring utilities. Elasticksearch and Kibana were installed on the Ubuntu Server1 while Logstash server was installed on the CentOS. The good results of service and browser tests verified that the Elastic Stack components have been correctly configured and running without problem. The whole activity has proven that the log monitoring that comes with automation not only helps in server management, bug fixing, and system checks but also that everything happens with less effort and in a quicker manner.

## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

**Tools Used:** **Claude**

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

**Purpose of Use:** **For debugging errors of the codes**

*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

**Extent of Use:** **Some code doesnt run for me has errors on the vm like the path file of the server's managed node and wrong alignment syntax changes.**

*Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.*

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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